ELECTRONIC ANSWER BOOK

I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM

Instruction: Using Microsoft Word/WordPad/Notepad, type Student's name, Student ID number, type Chapter title and applicable Question Number for Assignment, Mid-Term Exam or Final Exam (as listed under Brightspace, Quizzes) before typing Solution below. One separate file for each Assignment/Mid-Term/Final Exam Question.

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Assignment - Chapter title - applicable Question number:
Assignment – SOM (4%) – Question 1

Solution:

$\alpha = 7; S = 50; M = 5; a = 2;\ b = 0.06;\ c = -0.04;$

The lateral feedback weights of the Mexican hat function are given by:

$f_{-2j} = f_{-1j} = f_{0j} = f_{1j} = f_{2j} = 0.06$ for $j = 1$ to 11
$f_{-5j} = f_{-4j} = f_{-3j} = f_{3j} = f_{4j} = f_{5j} = -0.04$ for $j = 1$ to 11

	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	y_{10}	y_{11}
$k-1$	0.0000	0.1167	0.3455	0.6545	0.9045	1.0336	0.9045	0.6545	0.3455	0.1167	0.0000

Computation of $y_j(k)$ for $j = 1$ to 11 at any $k \geq 1$ is

$$y_j(k) = F[7 \sum_{m=-5}^{5} f_{mj} y_{j+m}(k-1)]$$

$$\begin{aligned} = F[7\{&f_{-5j}y_{j-5}(k-1) + f_{-4j}y_{j-4}(k-1) + f_{-3j}y_{j-3}(k-1) + f_{-2j}y_{j-2}(k-1) \\ &+ f_{-1j}y_{j-1}(k-1) + f_{0j}y_j(k-1) + f_{1j}y_{j+1}(k-1) + f_{2j}y_{j+2}(k-1) \\ &+ f_{3j}y_{j+3}(k-1) + f_{4j}y_{j+4}(k-1) + f_{5j}y_{j+5}(k-1)\}] \end{aligned}$$

For k, and j = 1, we have

$$\mathbf{y_1}(k) = F[7\sum_{m=-5}^{5} f_{m,1} y_{1+m}(k-1)]$$

$$= F[7\{-0.04 \times (0.0) - 0.04 \times (0.0) - 0.04 \times (0.0) + 0.06 \times (0.0) + 0.06 \times (0.0) + 0.06 \times (0.0000) + 0.06 \times (0.1167) + 0.06 \times (0.3455) - 0.04 \times (0.6545) - 0.04 \times (0.9045) - 0.04 \times (1.0336)\}]$$

$$= F[-0.5318]$$

$$\mathbf{y_1(k) = 0}$$

For k, and j = 2, we have

$$\mathbf{y_2}(k) = F[7\sum_{m=-5}^{5} f_{m,2} y_{2+m}(k-1)]$$

$$= F[7\{-0.04 \times (0.0) - 0.04 \times (0.0) - 0.04 \times (0.0) + 0.06 \times (0.0) + 0.06 \times (0.0000) + 0.06 \times (0.1167) + 0.06 \times (0.3455) + 0.06 \times (0.6545) - 0.04 \times (0.9045) - 0.04 \times (1.0336) - 0.04 \times (0.9045)\}]$$

$$= F[-0.3269]$$

$$\mathbf{y_2(k) = 0}$$

For k, and j = 3, we have

$$\mathbf{y_3}(k) = F[7\sum_{m=-5}^{5} f_{m,3} y_{3+m}(k-1)]$$

$$= F[7\{-0.04 \times (0.0) - 0.04 \times (0.0) - 0.04 \times (0.0) + 0.06 \times (0.0000) + 0.06 \times (0.1167) + 0.06 \times (0.3455) + 0.06 \times (0.6545) + 0.06 \times (0.9045) - 0.04 \times (1.0336) - 0.04 \times (0.9045) - 0.04 \times (0.6545)\}]$$

$$= F[0.1230]$$

$$\mathbf{y_3(k) = 0.1230}$$

For k, and j = 4, we have

$$\mathbf{y_4}(\mathrm{k}) = \mathrm{F}[7 \sum_{\mathrm{m}=-5}^{5} \mathrm{f}_{\mathrm{m},4} \mathrm{y}_{4+\mathrm{m}}(\mathrm{k}-1)]$$

$$\begin{aligned} = \mathrm{F}[7\{&-0.04 \times (0.0) - 0.04 \times (0.0) - 0.04 \times (0.0000) + 0.06 \times (0.1167) \\ &+ 0.06 \times (0.3455) + 0.06 \times (0.6545) + 0.06 \times (0.9045) \\ &+ 0.06 \times (1.0336) - 0.04 \times (0.9045) - 0.04 \times (0.6545) \\ &- 0.04 \times (0.3455)\}] \end{aligned}$$

$$= \mathrm{F}[0.7498]$$

$$\mathbf{y_4(k) = 0.7498}$$

For k, and j = 5, we have

$$\mathbf{y_5}(\mathrm{k}) = \mathrm{F}[7 \sum_{\mathrm{m}=-5}^{5} \mathrm{f}_{\mathrm{m},5} \mathrm{y}_{5+\mathrm{m}}(\mathrm{k}-1)]$$

$$\begin{aligned} = \mathrm{F}[7\{&-0.04 \times (0.0) - 0.04 \times (0.0000) - 0.04 \times (0.1167) + 0.06 \times (0.3455) \\ &+ 0.06 \times (0.6545) + 0.06 \times (0.9045) + 0.06 \times (1.0336) \\ &+ 0.06 \times (0.9045) - 0.04 \times (0.6545) - 0.04 \times (0.3455) \\ &- 0.04 \times (0.1167)\}] \end{aligned}$$

$$= \mathrm{F}[1.2685]$$

$$\mathbf{y_5(k) = 1.2685}$$

For k, and j = 6, we have

$$\mathbf{y_6}(\mathrm{k}) = \mathrm{F}[7 \sum_{\mathrm{m}=-5}^{5} \mathrm{f}_{\mathrm{m},6} \mathrm{y}_{6+\mathrm{m}}(\mathrm{k}-1)]$$

$$\begin{aligned} = \mathrm{F}[7\{&-0.04 \times (0.0000) - 0.04 \times (0.1167) - 0.04 \times (0.3455) + 0.06 \times (0.6545) \\ &+ 0.06 \times (0.9045) + 0.06 \times (1.0336) + 0.06 \times (0.9045) \\ &+ 0.06 \times (0.6545) - 0.04 \times (0.3455) - 0.04 \times (0.1167) \\ &- 0.04 \times (0.0000)\}] \end{aligned}$$

$$= \mathrm{F}[1.4848]$$

$$\mathbf{y_6(k) = 1.4848}$$

For k, and j = 7, we have

$$\mathbf{y_7}(k) = F[7 \sum_{m=-5}^{5} f_{m,7} y_{7+m}(k-1)]$$

$$\begin{aligned} = F[7\{&-0.04 \times (0.1167) - 0.04 \times (0.3455) - 0.04 \times (0.6545) + 0.06 \times (0.9045) \\ &+ 0.06 \times (1.0336) + 0.06 \times (0.9045) + 0.06 \times (0.6545) \\ &+ 0.06 \times (0.3455) - 0.04 \times (0.1167) - 0.04 \times (0.0000) - 0.04 \times (0.0)\}] \end{aligned}$$

$$= F[1.2685]$$

$$\mathbf{y_7(k) = 1.2685}$$

For k, and j = 8, we have

$$\mathbf{y_8}(k) = F[7 \sum_{m=-5}^{5} f_{m,8} y_{8+m}(k-1)]$$

$$\begin{aligned} = F[7\{&-0.04 \times (0.3455) - 0.04 \times (0.6545) - 0.04 \times (0.9045) + 0.06 \times (1.0336) \\ &+ 0.06 \times (0.9045) + 0.06 \times (0.6545) + 0.06 \times (0.3455) \\ &+ 0.06 \times (0.1167) - 0.04 \times (0.0000) - 0.04 \times (0.0) - 0.04 \times (0.0)\}] \end{aligned}$$

$$= F[0.7498]$$

$$\mathbf{y_8(k) = 0.7498}$$

For k, and j = 9, we have

$$\mathbf{y_9}(k) = F[7 \sum_{m=-5}^{5} f_{m,9} y_{9+m}(k-1)]$$

$$\begin{aligned} = F[7\{&-0.04 \times (0.6545) - 0.04 \times (0.9045) - 0.04 \times (1.0336) + 0.06 \times (0.9045) \\ &+ 0.06 \times (0.6545) + 0.06 \times (0.3455) + 0.06 \times (0.1167) \\ &+ 0.06 \times (0.0000) - 0.04 \times (0.0) - 0.04 \times (0.0) - 0.04 \times (0.0)\}] \end{aligned}$$

$$= F[0.1230]$$

$$\mathbf{y_9(k) = 0.1230}$$

For k, and $j = 10$, we have

$$\mathbf{y_{10}}(k) = F[7 \sum_{m=-5}^{5} f_{m,10} y_{10+m}(k-1)]$$

$$= F[7\{-0.04 \times (0.9045) - 0.04 \times (1.0336) - 0.04 \times (0.9045) + 0.06 \times (0.6545) + 0.06 \times (0.3455) + 0.06 \times (0.1167) + 0.06 \times (0.0000) + 0.06 \times (0.0) - 0.04 \times (0.0) - 0.04 \times (0.0) - 0.04 \times (0.0)\}]$$

$$= F[-0.3269]$$

$$\mathbf{y_{10}(k) = 0}$$

For k, and $j = 11$, we have

$$\mathbf{y_{11}}(k) = F[7 \sum_{m=-5}^{5} f_{m,11} y_{11+m}(k-1)]$$

$$= F[7\{-0.04 \times (1.0336) - 0.04 \times (0.9045) - 0.04 \times (0.6545) + 0.06 \times (0.3455) + 0.06 \times (0.1167) + 0.06 \times (0.0000) + 0.06 \times (0.0) + 0.06 \times (0.0) - 0.04 \times (0.0) - 0.04 \times (0.0) - 0.04 \times (0.0)\}]$$

$$= F[-0.5318]$$

$$\mathbf{y_{11}(k) = 0}$$

Therefore, the values of the outputs y_1 to y_{11} at iteration k are:

	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	y_{10}	y_{11}
$k-1$	0.0000	0.1167	0.3455	0.6545	0.9045	1.0336	0.9045	0.6545	0.3455	0.1167	0.0000
k	0.0000	0.0000	0.1230	0.7498	1.2685	1.4848	1.2685	0.7498	0.1230	0.0000	0.0000

The maximum output at iteration k is $\mathbf{y_6 = 1.4848}$
